

Importing and Turtle



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In today's class, we are going to be taking a short break from the fundamentals of programming to explore something more fun and interactive!



So far, our programs have been written in one file

This is not super realistic as programs often consist of hundreds, thousands, and even millions of lines of code!



Too much code!

It would be almost impossible to maintain a codebase that big in one huge file, *especially* if multiple people were working on it together!

It makes it very difficult to share / reuse code since we would need to constantly copy-paste it into every new project we have!



To fix these issues, we need a way to split our programs up across multiple files and somehow reference them in one another!

`import` Statements

`import` statements allow us to tell Python to include code from other files in our code when we execute our programs!

With them, you can split your own programs into multiple, more focused, files and it allows you to reuse those files by referencing them in future programs as a `library`.

Additionally, you can also use `libraries` written by other people!

Python is actually *very* good at this and it is why it is so commonly used in web development, data science, AI, and so many other areas!

import Statements in Action

We can reference things from these external files by **qualifying** them with the name of the file (without `.py`) followed by a period and the thing we want to access.

```
1 # Here we ask Python to 'include' the built-in math library.
2 import math # We don't add a ".py" here!
3
4 print(f"e is equal to {math.e}")
5
6 degrees: float = 270.0
7 print(f"{degrees}° in radians is {math.radians(degrees) / math.pi}π")
8
9 a: int = 30
10 b: int = 20
11 print(f"The gcd of {a} and {b} is {math.gcd(a, b)}")
12 print(f"The lcm of {a} and {b} is {math.lcm(a, b)}")
```

```
e is equal to 2.718281828459045
270.0° in radians is 1.5π
The gcd of 30 and 20 is 10
The lcm of 30 and 20 is 60
```

Now, let's look at the `turtle` library!

Turtle is a built-in graphics library that allows us to draw things to the screen!

